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fairmetrics: An R package for group fairness evaluation

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Summary

Fairness is a growing area of machine learning (ML) that focuses on ensuring that models do not produce systematically biased outcomes across groups defined by protected attributes, such as race, gender, or age. The `fairmetrics` R package provides a user-friendly framework for rigorously evaluating group-based fairness criteria, including independence (e.g., statistical parity), separation (e.g., equalized odds), and sufficiency (e.g., predictive parity) for binary protected attributes. The package provides both point and interval estimates for a variety of commonly used criteria. `fairmetrics` also includes an example dataset derived from the Medical Information Mart for Intensive Care, version II (MIMIC-II) database (Goldberger et al., 2000; J. Raffa, 2016; J. D. Raffa et al., 2016) to demonstrate its use.

Statement of Need

ML models are increasingly used in high-stakes domains such as criminal justice, healthcare, finance, employment, and education (Gao et al., 2024; Mattu, 2016; Mehrabi et al., 2021). Existing fairness evaluation software report point estimates and/or visualizations, without any measures of uncertainty. This limits users' ability to determine whether observed disparities are statistically significant. `fairmetrics` addresses this limitation by including confidence intervals for both difference and ratio based fairness metrics to enable more robust and statistically grounded fairness assessments.

Fairness Criteria

`fairmetrics` is designed to evaluate fairness of binary classification models across binary protected attributes. The package supports the evaluation of metrics belonging to three major group fairness criteria:

- **Independence:** Statistical Parity (compares the overall rate of positive predictions between groups).
- **Separation:** Equal Opportunity (compares false negative rates between groups), Predictive Equality (compares false positive rates between groups), Balance for Positive Class (compares the average predicted probabilities among individuals whose true outcome is positive across groups), and Balance for Negative Class (compares the average predicted probabilities among individuals whose true outcome is negative across groups).
- **Sufficiency:** Positive Predictive Parity (compares the positive predictive values across groups), Negative Predictive Parity (compares the negative predictive values across groups).

The package also includes additional metrics, such as the Brier Score Parity (compares the Brier score across groups), Accuracy Parity (compares the overall accuracy across groups), and Treatment Equality (compares the ratio of false negatives to false positives across groups).

Evaluating Fairness Criteria

The input required to evaluate model fairness with the `fairmetrics` package is a `data.frame` or `tibble` containing the model's predicted probabilities, the true outcomes, and the protected attribute. Figure 1 shows the workflow for using `fairmetrics`.

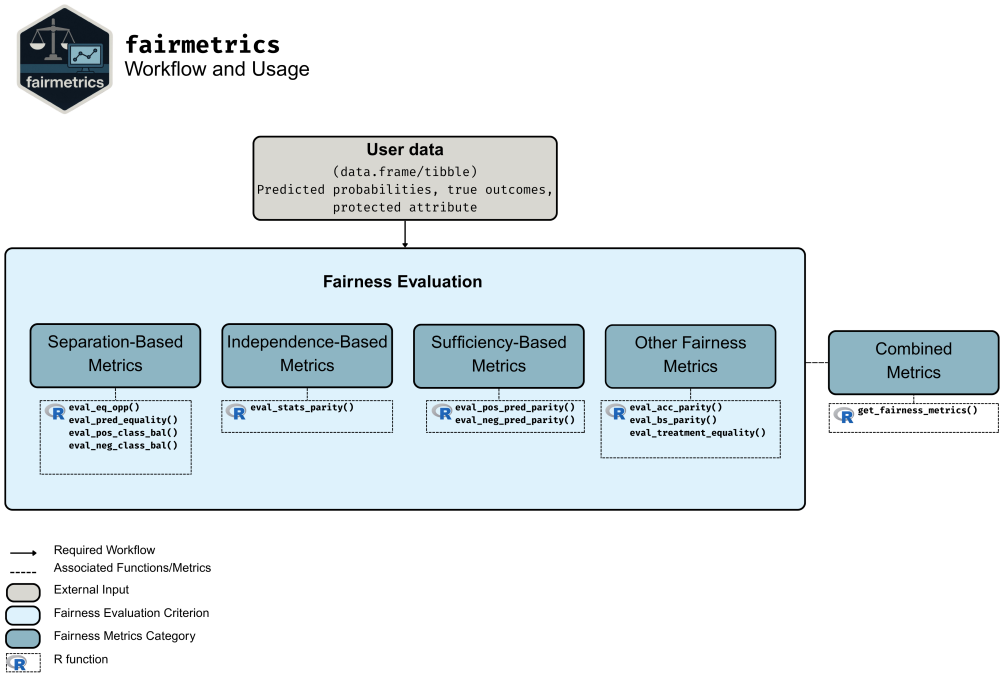

Figure 1: Workflow for using `fairmetrics` to evaluate model fairness across multiple criteria.

A simple example of how to use the `fairmetrics` package is illustrated below. The example makes use of the `mimic_preprocessed` dataset, a pre-processed version of the Indwelling Arterial Catheter (IAC) Clinical dataset, from the MIMIC-II clinical database (Goldberger et al., 2000; J. Raffa, 2016; J. D. Raffa et al., 2016).

While the choice of fairness metric used is context dependent, we show all criteria available with the `get_fairness_metrics()` function for illustrative purposes. In this example, we evaluate the model's fairness with respect to the binary protected attribute `gender`. The model is trained on a subset of the data and the predictions are made and evaluated on a test set. A statistically significant difference across groups at a given level of significance is indicated when the confidence interval for a difference-based metric does not include zero or when the interval for a ratio-based metric does not include one.

```
# Train a classification model (e.g., random forest).
# Add the vector of predicted probabilities to the test data
# to evaluate fairness.
library(fairmetrics)
# Setting alpha=0.05 for 95% confidence intervals
get_fairness_metrics(
```

```
  data = test_data,
  outcome = "day_28_flg",
  group = "gender",
  probs = "pred",
  cutoff = 0.41,
  alpha = 0.05
)
```

```
           Fairness Assessment                                Metric
1          Statistical Parity                Positive Prediction Rate
2           Equal Opportunity                     False Negative Rate
3         Predictive Equality                     False Positive Rate
4  Balance for Positive Class           Avg. Predicted Positive Prob.
5  Balance for Negative Class           Avg. Predicted Negative Prob.
6  Positive Predictive Parity               Positive Predictive Value
7  Negative Predictive Parity               Negative Predictive Value
8          Brier Score Parity                            Brier Score
9     Overall Accuracy Parity                               Accuracy
10         Treatment Equality (False Negative)/(False Positive) Ratio

   GroupFemale GroupMale Difference    95% Diff CI Ratio 95% Ratio CI
1         0.17      0.08       0.09   [0.05, 0.13]  2.12 [1.49, 3.04]
2         0.38      0.62      -0.24 [-0.39, -0.09]  0.61 [0.44, 0.86]
3         0.08      0.03       0.05   [0.02, 0.08]  2.67  [1.4, 5.08]
4         0.46      0.37       0.09   [0.04, 0.14]  1.24 [1.09, 1.42]
5         0.15      0.10       0.05   [0.03, 0.07]  1.50 [1.29, 1.74]
6         0.62      0.66      -0.04  [-0.21, 0.13]  0.94 [0.72, 1.22]
7         0.92      0.90       0.02  [-0.02, 0.06]  1.02 [0.98, 1.07]
8         0.09      0.08       0.01  [-0.01, 0.03]  1.12 [0.89, 1.43]
9         0.87      0.88      -0.01  [-0.05, 0.03]  0.99 [0.94, 1.04]
10        1.03      3.24      -2.21 [-4.38, -0.04]  0.32 [0.15, 0.68]
```

Users can also compute individual metrics using functions like `eval_eq_opp()` to test specific fairness conditions. Full usage examples are provided in the package documentation.

Related Work

Other R packages similar to `fairmetrics` include `fairness` (Kozodoi & V. Varga, 2021), `fairmodels` (Wiśniewski & Biecek, 2022) and `mlr3fairness` (Pfisterer et al., 2024). `fairmetrics` differs from these packages in two ways. The first difference is that `fairmetrics` calculates ratio and difference-based group fairness metrics and their corresponding confidence intervals, allowing for more meaningful inferences about the fairness criteria. The second difference is that `fairmetrics` does not possess any external dependencies and has a lower memory footprint. Table 1 shows the comparison of memory used and dependencies required when loading each library.

Package	Memory (MB)	Dependencies
fairmodels	17.02	29
fairness	117.61	141
mlr3fairness	58.11	45
fairmetrics	0.05	0

Table 1: Memory usage (in MB) and dependencies of 'fairmetrics' vs similar packages.

For Python users, the `fairlearn` library (Weerts et al., 2023) provides additional fairness

metrics and algorithms. The `fairmetrics` package is designed for seamless integration with R workflows, making it a more convenient choice for R users.

Licensing and Availability

The `fairmetrics` package is under the MIT license. It is available on CRAN and can be installed by using `install.packages("fairmetrics")`. Full documentation and its examples are available at: https://jianhuig.github.io/fairmetrics/articles/fairmetrics.html. Source code and issue tracking are hosted on GitHub: https://github.com/jianhuig/fairmetrics/.

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